This allows them to fly in Ready To Fly Mode. Almost, all latest drones can use 2.4GHz or 5.8GHz as operating frequencies.

Latest drones have intelligent flight controllers and modes such as Follow Me, Waypoints, Active Tracking, Return To Home and others for a controlled drive. Gimbal technology allows for any vibration from the drones to not reach the camera. It allows tilting the camera while in flight, creating unique angles required for aerial photography, film or 3D imagery. Many are three-axis stabilised gimbals with two working modes—non-FPV mode and FPV mode—while some have 360-degree stabilisation. ActiveTrack technology enables the drone pilot to select another moving object, for example, a car, cyclist or another drone, and to follow it without assistance from a tracker chip or beam.

Water drones like PowerDolphin can find fish, lure it and perform troll fishing. Real-time detection underwater helps anglers accurately determine active fishing spots. Such drones are equipped with an intelligent sonar device and GPS waypoint function to draw underwater topographic maps. Underwater drones with 4k camera can dive up to 150 metres and are perfect for professional underwater filming. Also incorporated in the design are self-balancing and anti-current algorithms for best underwater performance.

The future of drone technology

Smart drones like Solo have more efficient motors, more accurate sensors, better onboard processors and software, advanced autopilot system and built-in compliance technology for safe, effective flight control, for new opportunities in transport and logistics. Smart sensors control and monitor the flight. These are guided by computer vision systems along with object detection and collision avoidance features. New forms of AI or algorithms make these even more adaptable.

Totally autonomous drones constructed of high-impact plastic, which makes them lighter and more robust, need no input or control from the pilot. These drones deliver data in real time once the destination or task is set. These can monitor fuel levels and watch out for damage to their propellers, cameras and sensors. Lightweight, long-lasting battery or power source is needed for such drones and other UAVs to enable them to fly for hours. LightCense is a new technology that uses visible colour blink sequence to create an identification system for drones. The code can be identified through smartphones using an app.

A self-monitoring system is important for the commercial viability of drones for the automatic delivery of goods or emergency services. This helps make sure that packages actually arrive at their intended destination. Industrial inspection, delivery, surveillance and agricultural applications are likely to be among the first commercial drone applications. EFY